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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution Affiliated to VTU)
IV Semester B. E. Examinations Sept./Oct. – 2024
Industrial Engineering and Management
STATISTICS FOR DATA ANALYTICS

Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables and formula handbook permitted.

PART-A

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1	1.1	Give an example for ordinal data type.	01	1	1
	1.2	Give one distinction between Histogram and Box plot.	01	1	1
	1.3	What are mutually exclusive events?	01	2	1
	1.4	State Baye's Theorem.	01	2	1
	1.5	Define Random variable.	01	1	2
	1.6	What is a probability mass function?	01	2	2
	1.7	Sketch the shape of a discrete uniform distribution.	01	1	1
	1.8	Define a Poisson random variable.	01	2	1
	1.9	What is the theoretical range of a Normal distribution?	01	1	2
	1.10	What is a Z score?	01	2	2
	1.11	State the central limit theorem.	01	1	3
	1.12	Sketch the relative spread of a population which is normally distributed and the sampling distribution of means. (Mean μ standard deviation σ sample size n).	01	2	3
	1.13	What is the meaning of an Empirical model?	01	2	3
	1.14	List any one of the properties of Least square estimation.	01	1	3
	1.15	Define correlation coefficient.	01	1	4
	1.16	Mention any one of the common abuses of regression.	01	2	4
	1.17	Give the correct structure of a pair of hypothesis for any one of the population parameter.	01	1	4
	1.18	Define prediction interval.	01	1	4
	1.19	What is the meaning of the term Goodness of fit?	01	2	4
	1.20	When do you use the F statistic?	01	2	4

PART-B

2	a	Given below is a frequency distribution of the length of phone calls made by 175 people during a weekend. Construct a histogram for these data																					
		<table><tr><td><i>Length in minutes</i></td><td>1-7</td><td>8-14</td><td>15-21</td><td>22-28</td><td>29-35</td><td>36-42</td><td>43-49</td><td>50-56</td></tr><tr><td><i>Frequency</i></td><td>45</td><td>32</td><td>34</td><td>22</td><td>16</td><td>12</td><td>9</td><td>5</td></tr></table>	<i>Length in minutes</i>	1-7	8-14	15-21	22-28	29-35	36-42	43-49	50-56	<i>Frequency</i>	45	32	34	22	16	12	9	5			
<i>Length in minutes</i>	1-7	8-14	15-21	22-28	29-35	36-42	43-49	50-56															
<i>Frequency</i>	45	32	34	22	16	12	9	5															
		i) Describe the general shape of the histogram. Does these appear to be a pattern? ii) Suppose all the people were making their calls from a room that had 10 different phones and each person knew which time class the call would belong to. Suggest an ordering so that all calls can be completed as fast as possible.	08	3	1																		

b	The university's library has been randomly surveying patrons over the last month to see who is using the library and what services they have been using. Patrons are classified as undergraduate student graduate or faculty. Services are classified as reference periodicals or books. The data for 350 people are given below. Assume a patron uses only one service per visit. <table><tr><td>Patron</td><td>Reference</td><td>Periodicals</td><td>Books</td></tr><tr><td>Undergraduate</td><td>44</td><td>26</td><td>72</td></tr><tr><td>Graduate</td><td>24</td><td>61</td><td>20</td></tr><tr><td>Faculty</td><td>16</td><td>69</td><td>18</td></tr><tr><td></td><td>84</td><td>156</td><td>110</td></tr></table> Find the probability that a randomly chosen patron i) Is a graduate student. ii) Visited the periodicals section, given that the patron is a graduate student. iii) Is a faculty member given a reference section visit? iv) Is an undergraduate who visited the book section?	Patron	Reference	Periodicals	Books	Undergraduate	44	26	72	Graduate	24	61	20	Faculty	16	69	18		84	156	110	08	2	1
Patron	Reference	Periodicals	Books																					
Undergraduate	44	26	72																					
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3	a The probability is 0.80 that a randomly selected technician will finish his or her project successfully. Let X be the number of technicians among a randomly selected group of 10 technicians who will finish their project successfully. i) Find the probability distribution of the random variable X. ii) Find the probabilities that at least three technicians will finish their projects successfully. iii) Find the probabilities that at most five technicians will finish their projects successfully. iv) Find the probability that between four and six (inclusive) technicians will finish their project successfully. b A total of 75 parts are received from the suppliers. We are informed that 8 defective parts were shipped by mistake and 5 parts have already been installed on machines. i) What is the probability that exactly 1 defective part was installed on a machine? ii) What is the probability of finding less than 2 defective parts? OR	08	3	2																				
4	a The number of surface flaws in plastic panels used in the interior of auto mobiles has a Poisson distribution with a mean of 0.05 flaws per square foot of plastic panel. Assume an automobile interior contains 10 square feet of plastic panel. i) What is the probability that there are no surface flaws in auto's interior? ii) If 10 cars are sold to a rental company, what is the probability that none of the 10 cars has any surface flaws? iii) If flaws 10 cars are sold to a rental company, what is the probability that at most one car has any surface flaws? b Let X denotes the number of bits received in error in a digital communication channel assume that X is a binomial random variable with variable with $P' = 0.001$. If 1000 sets are transmitted, determine the following: (i) $P(X = 1)$ (ii) $P(X \geq 1)$ (iii) $P(X \leq 2)$ (iv) Mean and Variance of X	08	3	3																				

5	a	Suppose the time it takes a data collection operator to fill out an electronic form for a database is uniformly distributed between 1.5 and 2.2 mins. i) What is the mean and variance of the time it takes an operator to fill out the form? ii) What is the probability that it will take less than two minutes to fill out the form?	08	4	3																												
	b	The reaction time of a drives to visual stimulus is normally distributed with a mean of 0.4 seconds and a standard deviation of 0.05 seconds. i) What is the probability that a reaction requires more than 0.5 seconds? ii) What is the probability that a reaction requires between 0.4 and 0.5 seconds? iii) What is the reaction time that is exceeded 90% of the time?	08	4	3																												
OR																																	
6	a	The thickness of the flange on an aircraft component is uniformly distributed between 0.95 and 1.05 millimeters. i) Determine the cumulative distribution function of flange thickness. ii) Determine the proportion of flanges that exceed 1.02 millimeters. iii) What thickness is exceeded by 90% of the flanges? iv) Determine the mean and variance of the flange thickness.	08	3	3																												
	b	The sick leave time of employees in a firm in a month is normally distributed with a mean of 100 hours and a standard deviation of 20 hours. i) What is the probability that the sick leave time for the next month will be between 50 and 80 hours? ii) How much time should be budgeted for sick leave if the budgeted amount should be exceeded with a probability of only 10%?	08	4	2																												
7	a	Gasoline consumption of a test automobile was recorded at speeds (x) ranging from 56 to 112 km/hr. The observed gasoline consumption were converted to distance travelled per liter of gas line (y). The following information was compiled $\eta = 12, \Sigma x = 964, \Sigma x^2 = 84116, \Sigma xy = 13418.4, \Sigma y = 165.3, \Sigma y^2 = 2282.45, \bar{x} = 80.333, \bar{y} = 13.755$ i) Find the regression equation for y on x . ii) Find the correlation coefficient.	12	4	3																												
	b	Give the benefits of initial visualization of data pattern using a scatter plot. Give an example to highlight the visualization of patterns in data.	04	2	3																												
OR																																	
8	a	For a sample of 12 trees the volume of lumber ($in\ m^3$) and diameter ($in\ cm$) at a fixed height above ground level was measured. The results were as follows <table border="1"><tr><td>Diameter</td><td>35.1</td><td>48.4</td><td>47.9</td><td>35.3</td><td>47.3</td><td>26.4</td><td>33.8</td><td>45.3</td><td>25.2</td></tr><tr><td>Volume</td><td>0.81</td><td>1.39</td><td>1.31</td><td>0.67</td><td>1.46</td><td>0.47</td><td>0.8</td><td>1.69</td><td>0.3</td></tr></table> <table border="1"><tr><td>Diameter</td><td>28.5</td><td>30.1</td><td>30</td></tr><tr><td>Volume</td><td>0.19</td><td>0.63</td><td>0.64</td></tr></table>	Diameter	35.1	48.4	47.9	35.3	47.3	26.4	33.8	45.3	25.2	Volume	0.81	1.39	1.31	0.67	1.46	0.47	0.8	1.69	0.3	Diameter	28.5	30.1	30	Volume	0.19	0.63	0.64			
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		i) Construct a scatter plot of volume (y) versus diameter (x) ii) Compute the parameters of the least square line for predicting volume from diameters iii) If two trees differ in diameter by 8 cm, how much would you predict this volumes to differ? iv) Predict the volume for a tree whose diameter 44 cm? v) For what diameter would a predict a volume of $1m^3$?	14 02	4 2	3 3
	b	How is causation different from correlation?			
9	a	A manufacture produces piston rings for an automobile engine. It is known that the ring diameter is normally distributed with $\sigma = 0.001$ $\bar{x} = 74.036$ millimeter based on a random sample of 15 rings. i) Construct a 99% two sided confidence interval on the mean piston ring diameter. ii) Construct a 99% lower confidence band on the mean piston ring diameter.	08	3	4
	b	A post mix beverage machine is adjusted to release a certain amount of syrup into a chamber, where it is mixed with carbonated water. A random sample of 25 beverages was found to have a mean syrup content of $\bar{x} = 1.10$ fluid ounces and a standard deviation of $s = 0.015$ fluid ounces. Find a 95% confidence interval on the mean volume of syrup dispensed.	08	3	4
		OR			
10	a	A consumer product company is formulating a new shampoo and is interested in foam height (in mm). Foam height is approximately normally distributed and has a standard deviation of 20 mm. The company worker to test $H_0: \mu = 175mm$ versus $H_1: \mu > 175mm$ using the results of $n = 10$ samples. Suppose that the sample data result in $\bar{x} = 190mm$ i) What conclusions would you reach? ii) How "unusual" is the sample value in mm if the true mean is really 175 mm? That is what is the probability that you would observe a sample average as large as 190 mm (or larger). If the true mean foam height was really 175 mm?	10	3	4
	b	Outline the 8 steps procedure for Testing of hypothesis. Give an example for a null hypothesis and alternate hypothesis.	06	4	4